

Formal Forward Coverage of Two Modulo Nine

Collatz proof project

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Abstract

We formalize Corollary 5.8 of Monks et al.: every positive natural Collatz orbit visits two modulo nine. Applying the result to every tail gives arbitrarily late visits and a conditional reduction of convergence to this progression. This is a Lean formalization of a known theorem; it does not exclude divergence or nontrivial cycles.

Write $T(n) = n/2$ for even n and $T(n) = (3n + 1)/2$ for odd n .

Theorem 1. *For every $n > 0$ and $K \geq 0$, some $t \geq K$ satisfies $T^t(n) \equiv 2 \pmod{9}$.*

Proof. Our existing formal arithmetic results give a tail outside multiples of three and an even value on every natural orbit. On the six remaining residues the possible transitions, with parity suppressed, are

r	1	2	4	5	7	8
$T(r)$	5, 2	1, 8	2	7, 8	8, 2	4, 8.

Assume a tail avoids residue two. The finite transition constraints force its value after three steps to have residue eight: residue four must be excluded since its successor is two, and the remaining non-eight chain is 1, 5, 7, 8. Applying this argument to every further tail shows that all values from time three onward have residue eight. An even value of residue eight has successor residue four, a contradiction. This proves a visit exists. Starting the same argument at $T^K(n) > 0$ gives the stated time bound. $\square$

The even-value lemma uses finite parity injectivity: if a natural seed n had only odd steps, its parity prefix of length $k = n + 2$ would agree with $2^k - 1$, forcing equality of their residues modulo 2^k . Both lie below 2^k , while $n < 2^k - 1$, a contradiction.

Convergence reduction and remaining gap. If every positive $n \equiv 2 \pmod{9}$ reaches one, then every positive seed reaches one by composition with the hitting time above. That premise is unproved. In particular, the result cannot be combined directly with the least-noncontracting-seed restriction: the property $2^t \leq 3^{j_t(n)}$ for all t is not supplied at the selected residue visit. Coverage and the existence of noncontracting tails are separate existential statements; their witnesses have not been shown to coincide.

Formal artifacts. `Collatz.ResidueCoverage` proves `exists_two_mod_nine`, `arbitrarily_late_two_mod_nine`, and `reaches_one_of_two_mod_nine`. The existing `exists_even_value` is now public. The proofs use exact arithmetic and existing parity results, with no imported literature axiom or finite seed cutoff.

References

- [1] K. Monks, K. G. Monks, K. M. Monks, and M. Monks, *Strongly sufficient sets and the distribution of arithmetic sequences in the $3x + 1$ graph*, arXiv:1204.3904v2 (2012), Corollary 5.8.
<https://arxiv.org/abs/1204.3904>.